# Re-imagining Mapping:

# Food Insecurity in New York State

Lu Yii Wong

Sustainable Urban Environments Capstone II

Professor David Holzman

3 May 2023

**Abstract**

Food insecurity is a national concern in the United States of America. In 2020, at least 13.8 million households in America reported experiencing food insecurity over a period of 12 months. The capstone project aims to identify a blueprint to build a robust food insecurity map that makes information on the topic accessible. This is executed by evaluating existing food insecurity maps, Map the Meal Gap by Feeding America and Hunger Solutions New York's Food Insecurity and Federal Nutrition Program Participation Map. The proposed food insecurity map is designed to educate laypeople, stakeholders, and policymakers about local food insecurity and provide emergency food resources available in the community. This project focuses on New York State as a model for displaying local-level food insecurity.

**Introduction**

Food insecurity is a national concern in the United States of America. The United States Department of Agriculture and Economic Research Service (USDA ERS) defined food insecurity as "the limited or uncertain availability of nutritionally adequate and safe foods, or limited or uncertain ability to acquire acceptable foods in socially acceptable ways" (USDA ERS 2022). Access to food is a fundamental human right that everyone should have. Yet, even in an economically developed country like America, many people still experience food insecurity. This results from a multitude of circumstances, such as the lack of physical access to food or the inability to afford food.

In 2020, national household food insecurity measures by the USDA showed that 10.5% of households experienced food insecurity (Coleman-Jensen et al. 2021). Similarly, 10.5% of

households, which accounted for approximately 13.8 million households, in New York State (NYS) reported experiencing low to very low food insecurity within that year (Coleman-Jensen et al. 2021). Despite the large number of food-insecure households in NYS, the proportion of food-insecure households has declined since the Great 2008 economic recession. Nationally, average low and very low food security rates declined from the twenty-first-century peak of 14.9% in 2011 to 10.5% in 2020 as seen in Figure 1 (USDA ERS 2022).

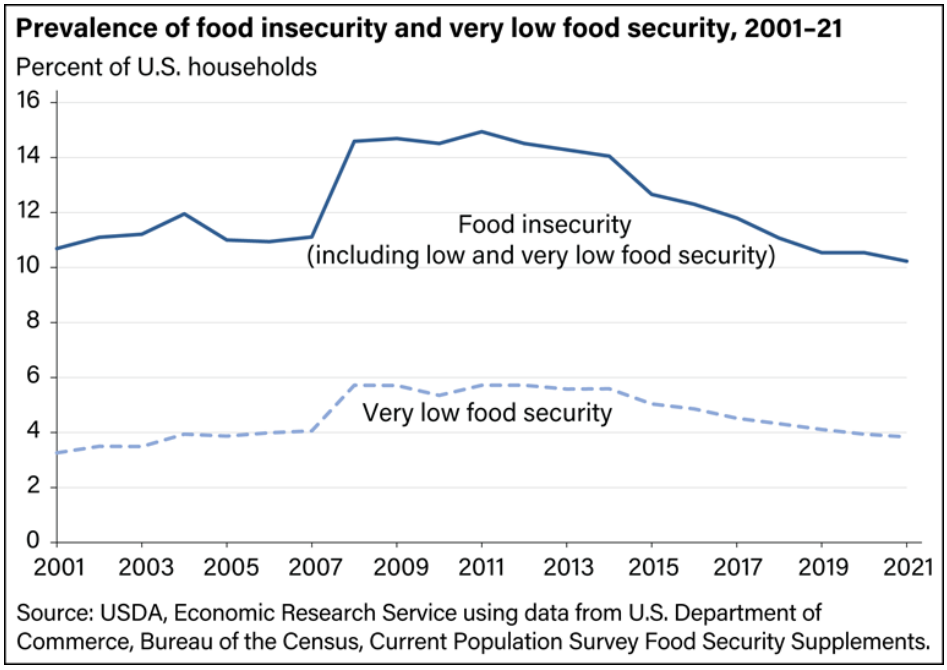


Figure 1: Graph depicting low and very low food security rates from 2001 to 2021 (USDA ERS 2022)

The impact and distribution of food insecurity experienced are not equal across all populations in the US. Although food insecurity has been a national concern for many decades, the COVID-19 pandemic shone new light and urgency on the issue. The COVID-19 pandemic exacerbated food insecurity among vulnerable populations. This included Black, Indigenous, and

People of Color (BIPOC) communities, households with children, and people whose jobs were impacted by the pandemic and economic recession (Clay and Rogus 2021). In 2020, the U.S. economy entered a recession, contracting dramatically by 31.2% in the second quarter (FRED, n.d.). The rise in unemployment rates during the early onset of the pandemic impacted many. It unearthed the people's reliance on local emergency food systems, such as food banks and soup kitchens.

Every state in the US is unique. Local characteristics of a state, such as its physical characteristics and political environment, impact how food insecurity is experienced and handled. Is there a way to predict how food insecurity solutions might impact levels of food insecurity in a county? I will examine existing food insecurity models, barriers, and solutions in this capstone. I will also propose a new food insecurity model that improves access to information and potentially predicts the growth or decline of food insecurity in New York State based on the solutions implemented. The model will be presented in the form of an interactive map. It will be designed for laypeople and policymakers to inform local food security knowledge and connect people to existing emergency food resources.

**Current food insecurity models**

Food insecurity is a challenging concept to quantify. The USDA quantifies food insecurity through an annual Current Population Survey (CPS) in the United States. The survey used by the USDA to assess household food security consists of 10 required questions for all households surveyed and eight additional questions for households with children under 18 years old (USDA 2022). Only households experiencing low and very low food insecurity are

considered food insecure. In this categorization, food insecurity ranges from least severe to most severe. Households are considered food insecure "if they report any combination of three or more conditions" preventing them from attaining adequate food to sustain themselves (USDA 2022). The most severe cases of household food insecurity are determined when respondents reported that "eating patterns for one or more household members were disrupted and their food intake reduced" because they did not have enough money or resources to get adequate food (USDA 2022).

Understanding how food insecurity is measured and defined in the US is crucial as it shapes how we perceive it. The food insecurity data collected from the CPS is the basis for most food insecurity models in the US. The existing framework may impact how resources to address food insecurity are prioritized. The main models I looked at for this project were Map the Meal Gap by Feeding America and Hunger Solutions New York's Food Insecurity and Federal Nutrition Program Participation Map.

Map the Meal Gap

Feeding America is the country's food bank network and a 501 (c)(3) nonprofit organization recognized by the Internal Revenue Service (IRS). The Feeding America organization has contributed significantly to the food security realm. Map the Meal Gap (MMG) is a comprehensive study on national food insecurity conducted by Feeding America to spread a greater understanding of food insecurity. The product of the study is a publicly-available interactive map that examines food insecurity rates and food costs on a local level, designed to help users understand this phenomenon. Resources like MMG are crucial tools that help spread

awareness of the severity of food insecurity. With heightened awareness, more attention can be "given to the role of systems and policies that keep certain households and communities food insecure, including discriminatory policies and systems" that exacerbate social and economic inequalities (Feeding America, n.d.).

The latest publication of MMG, released in 2022, "examines differences in local food insecurity rates by race and ethnicity" (Feeding America, n.d.). MMG has found a way to quantify and estimate the impacts of race and ethnicity on the impact on overall food insecurity rates. The populations identified in the estimates are "Black (all ethnicities), Latino (Hispanic), and white, non-Hispanic" populations (Map the Meal Gap 2022). Other BIPOC communities not included, such as Native Americans and Asian populations, are still disproportionately affected by food insecurity. MMG cites that these populations were excluded from current calculations because of small sample sizes and the lack of data availability (Map the Meal Gap 2022).

The MMG 2022 Technical Brief explains how race and ethnicity are disaggregated. Each race and ethnicity are separated into subgroups, and the "state-level relationships between food insecurity and subgroup determinants" are analyzed (Map the Meal Gap 2022). Subgroup-specific determinants would be unemployment, poverty, disability rate, homeownership rate, and the median income for the subgroup (Map the Meal Gap 2022). The estimates derived from this analysis are then applied at the county and congressional district levels. Including race and ethnicity in determining food insecurity rates give nuance to the issue. It highlights local food insecurity trends that impact the most vulnerable populations and might have otherwise been overlooked or obscured by data.

The 2022 MMG Technical Brief covers the methods employed to estimate food insecurity rates across all counties. Firstly, they determined state-level food insecurity using the following equation:

$$FI_{st} = \alpha + \beta_{UN}UN_{st} + \beta_{POV}POV_{st} + \beta_{MI}MI_{st} + \beta_{HISP}HISP_{st} + \beta_{BLACK}BLACK_{st} + \beta_{OWN}OWN_{st} + \beta_{DSBL}DSBL_{st} + \mu_t + \upsilon_s + \varepsilon_{st} \quad (1)$$

The above equation denotes that the state-level food insecurity rate is determined by using state-level data for the unemployment rate, poverty rate, median income, percentage of the Hispanic population, percentage of Black African-American population, percentage of homeowners, and percent of the disabled population (Map the Meal Gap 2022). The equation also accounts for errors, year-fixed, and state-fixed effects (Map the Meal Gap 2022).

Secondly, individual county-level and congressional district-level food insecurity rates are calculated using the following equation:

$$FI^*_c = \hat{\alpha} + \widehat{\beta_{UN}}UN_c + \widehat{\beta_{POV}}POV_c + \widehat{\beta_{MI}}MI_c + \widehat{\beta_{HISP}}HISP_c + \widehat{\beta_{BLACK}}BLACK_c + \widehat{\beta_{OWN}}OWN_C + \widehat{\beta_{DSBL}}DSBL_C + \widehat{\mu_{2018}} + \hat{v}_s \quad (2)$$

The same principles applied to the state-level equation are then applied to the county or congression district levels. An advantage of the MMG model is the representation of local-level information. “While [national-level data] can help tell a high-level story, experiences differ by location and community” (Hake, Engelhard, and Dewey 2022). Making distinctions on the local level is essential when employing local solutions. The MMG model can be considered one of the most comprehensive nationwide food insecurity maps since it considers various factors contributing to the severity of food insecurity experienced. MMG derives its data from the CPS, the American Community Survey (ACS), and the Bureau of Labor Statistics (BLS) (Map the

Meal Gap 2022). The model uses publicly available data and presents it in a clear and compelling visual manner. Figure 2 is an example of what the MMG interactive map looks like. From a user perspective, the site is relatively easy to navigate, and information is readily provided.

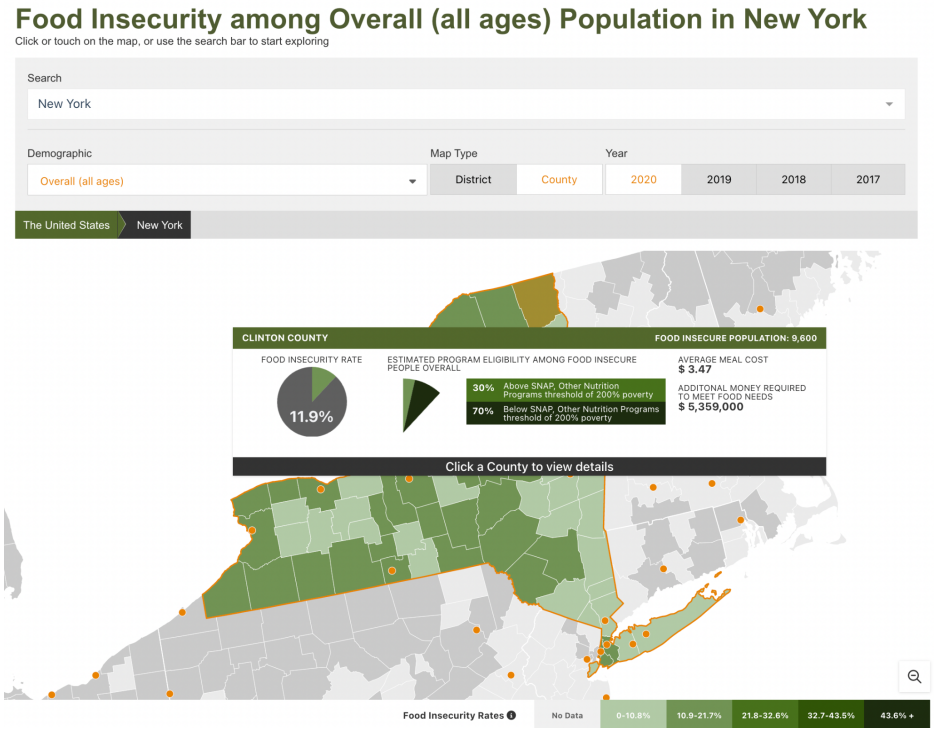


Figure 2: Example of MMG food insecurity map for New York State

<u>Hunger Solutions New York</u>

Hunger Solutions New York (HSNY) is a "statewide non-profit organization dedicated to alleviating hunger" (Hunger Solutions New York 2023). A significant part of the organization's work is to increase education on and participation in federal nutrition programs. HSNY published a New York State map that reports county-level participation in various federal nutrition programs. The program participation reported included "SNAP, WIC, Summer Food

Service Program, and CACFP Afterschool Meals" (Hunger Solutions New York, n.d.). This map comprises data from MMG, Food Research and Action Center (FRAC), NYS Department of Health WIC Program, and HSNY sources.

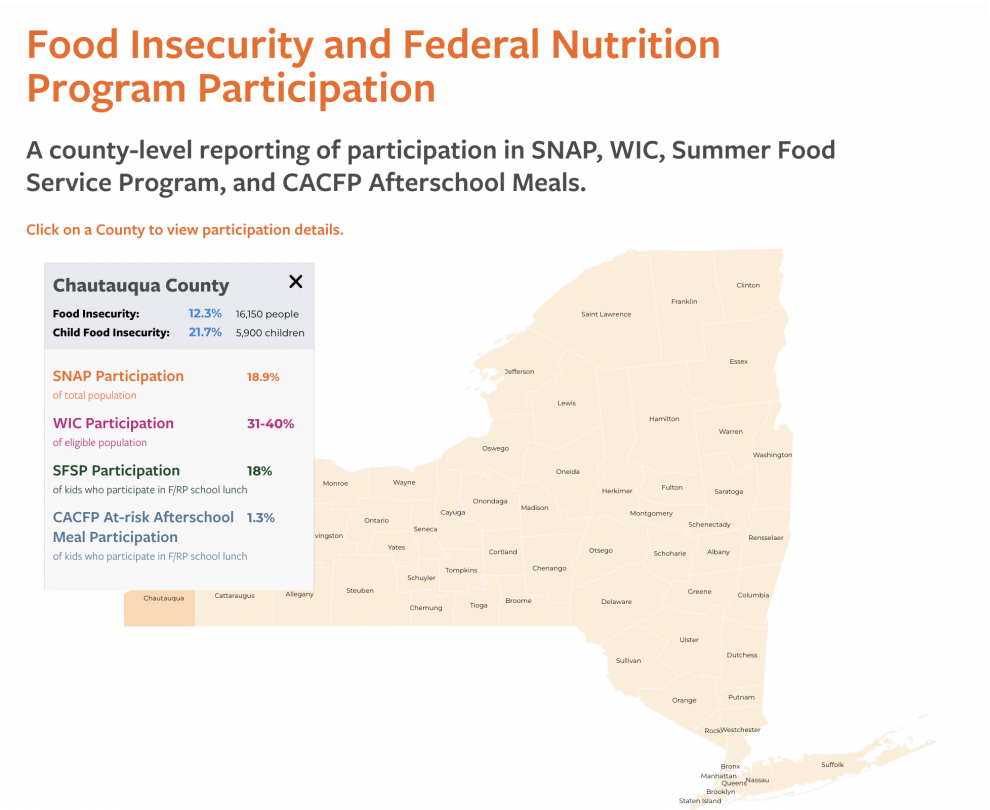


Figure 3: Example of HSNY food insecurity and nutrition program participation map

The HSNY map is a simplified map that displays important information on New York State by county. This example does not rework any raw data. Instead, the HSNY map identifies information that might be useful to help users understand the food insecurity issue in New York State. From a user's perspective, the map was straightforward to navigate. By selecting a county, the relevant information is highlighted in a popup window (Figure 3). A downside of this map is that does not accommodate county-to-county comparison. At a glance, the user cannot

differentiate between more and less food-insecure counties. Secondly, the year of the data used for food insecurity rate, SNAP participation, WIC participation, SFSP, and CACFP are from different years. This might impact the accuracy of any comparisons made.

**Comparing the models**

The MMG model is a more comprehensive and interactive visual tool to help users better understand food insecurity than the HSNY map. The MMG model allows users to click on specific states, congressional districts, and counties. Its pop-up tab lets users get a quick overview of the food insecurity statistics in each place. While the HSNY map has a similar feature, there is the absence of a color gradient feature that differentiates counties with higher or lower food insecurity rates. Yet, the HSNY map also has its advantages. This map includes a wide range of food security program participation rates. This provides essential information on the county's programs and the allocation of well-utilized food security resources. This information provides a clearer picture of New York State’s food landscape. In creating an informative visual tool, local context is vital to create tailored programs most suitable for a locality.

**Limitations of existing models**

Both MMG and HSNY maps provide essential information regarding food security on national and local levels respectively. Despite the strengths of these models, both tools have significant limitations. In these maps, minority populations are still undercounted and underrepresented in the measures of food insecurity. The foundational data used in both maps are from the MMG methods for measuring and defining food insecurity. As discussed earlier, the

MMG model weights food insecurity at the state and local levels using demographic data from the census. The MMG model accounts for disparities observed in two minority groups, Hispanic and Black African American populations. However, the American population is significantly more diverse. Existing literature acknowledges the disparities that all BIPOC populations experience in food security. While efforts are made through the MMG model to account for these disparities, other minority populations, such as Indigenous and Asian populations, are overlooked in current measures.

Food insecurity is prevalent in Native communities, with 25 percent of American Indians and Alaska Natives (AIAN) experiencing food insecurity consistently between 2000 and 2010, at a rate double of white Americans (Maillacheruvu 2022). The high proportion of AIAN food insecurity directly results from a long history of racist federal policies against Native communities, leaving many Native populations experiencing high food insecurity and poverty rates. A report published by the USDA found that the AIAN population was more likely to live in rural areas, with only 25.6 percent of individuals living in tribal areas within 1 mile of a supermarket, compared with 58.8 percent of all Americans (Kaufman, Dicken, and Williams 2014). The conversation around food security should also be rooted in social justice. Based on data published by the USDA, stark disparities in food security can be observed by race and ethnicity. In Figure 4, the graph shows that all racial minority groups, except for the Native Hawaiian, Asian American, or Pacific Islander (NHAAPI) population, consistently have more significant proportions of households experiencing food insecurity than White households. The statistics show apparent disparities and the lack of social justice in our food system.

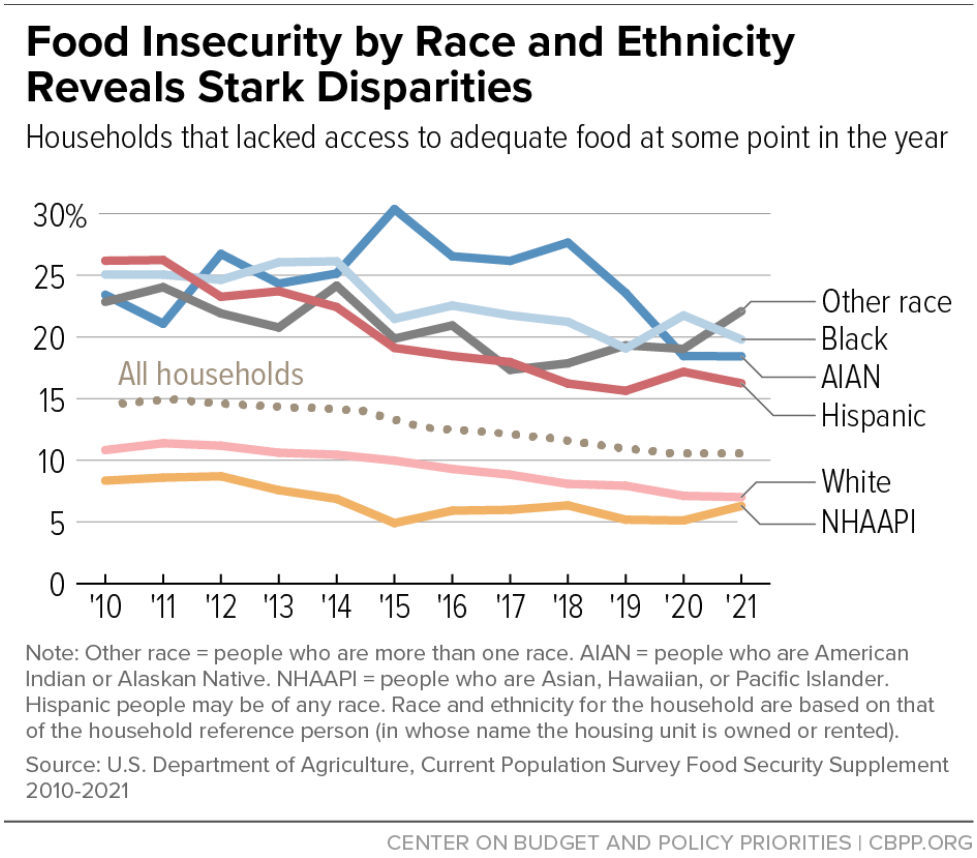


Figure 4: Food Insecurity Trends by Race and Ethnicity (Maillacheruvu 2022)

Despite such apparent disparities, why are these minority groups still underrepresented in our food insecurity measurements? In the MMG technical brief, the writers acknowledged that other minority groups apart from Black and Hispanic populations were "disproportionately affected by food insecurity; however, sample size and data availability have prevented [them] from including additional racial and/or ethnic representation in the model." (Map the Meal Gap 2022). This suggests that to improve our current food security models, we must expand our data collection methodology to increase the information gathered on minority populations. Filling the information gap can help us better understand which populations within a locality are impacted by food insecurity. This will improve the way in which we apply interventions and mitigate

factors that cause food insecurity. It will also allow us to tailor food security interventions that provide culturally appropriate and nutritious food to food-insecure communities rather than band-aid approaches of rationing “articles of subsistence” that only catered to providing adequate nutrition (Maillacheruvu 2022). I believe this will be a greater priority for states or counties with larger populations of underrepresented minority populations.

**Building An Improved Food Security Framework**

Exploring existing mapping tools for food security informed my experience in navigating food security in the US. As I learned more about food insecurity, I realized that existing tools and platforms catered to specific needs and often communicated selective information. I found that I had to rely on multiple platforms to gain a comprehensive understanding of the US food security landscape and purposefully seek out particular information. From the perspective of a layperson trying to educate themselves about food security, the current infrastructure has many barriers that make learning challenging. From the perspective of a person trying to access local resources for food assistance, finding clear information about programs and resources I might be eligible for was convoluted. This experience inspired me to design a single online resource that would concisely compile all essential information about food security and local food resources.

The purpose of the new tool is to inform laypeople and policymakers about the local food security landscape. Information consolidated on the platform aims to advocate for change in existing food policy by empowering people with accessible and transparent data. This proposed tool will also be modeled as a community asset that connects food-insecure residents to available food resources. These improvements will empower laypeople and policymakers to be better

informed about the local food landscape and increase local food resiliency through one consolidated platform.

New York State Food Map (NYS Food Map) is a visually appealing, interactive mapping model that I designed. The map is designed to be accessed online as a publicly available and free resource. The proposed model for NYS Food Map was created using Figma, a wireframing software, and Tableau, a visual analytics platform. Figure 5 is the landing page for the site. The landing page welcomes users and instructs “Learn more about food security in all New York State Counties through our interactive map. Click to play.” The design is kept simple to avoid distraction from the site's purpose while maintaining a casual tone to encourage users of all ages to interact with the model.

Figure 5: NYS Food Map Landing Page

When users click on the map, they will be brought to the main interactive interface. In Figure 6, the mapping tool displays food insecurity rates (FIR) in 2017 for all NYS counties. The site is thoughtfully designed to be accessible to all users, regardless of physical ability. Users can access auditory assistance and various language options on the bottom left of the page. The inclusion of accessible functionalities into the tool was prompted by personal experience navigating online resources. Existing platforms mainly serve English-speaking populations. Restricting resources to one language might exclude minority populations in the state that do not read English. Additionally, auditory assistive features are not frequently included in visual-oriented applications. While NYS Food Map is a visual and interactive tool, it offers value beyond data visualization. To maintain accessibility to all users, these functionalities were thoughtfully incorporated into the design.

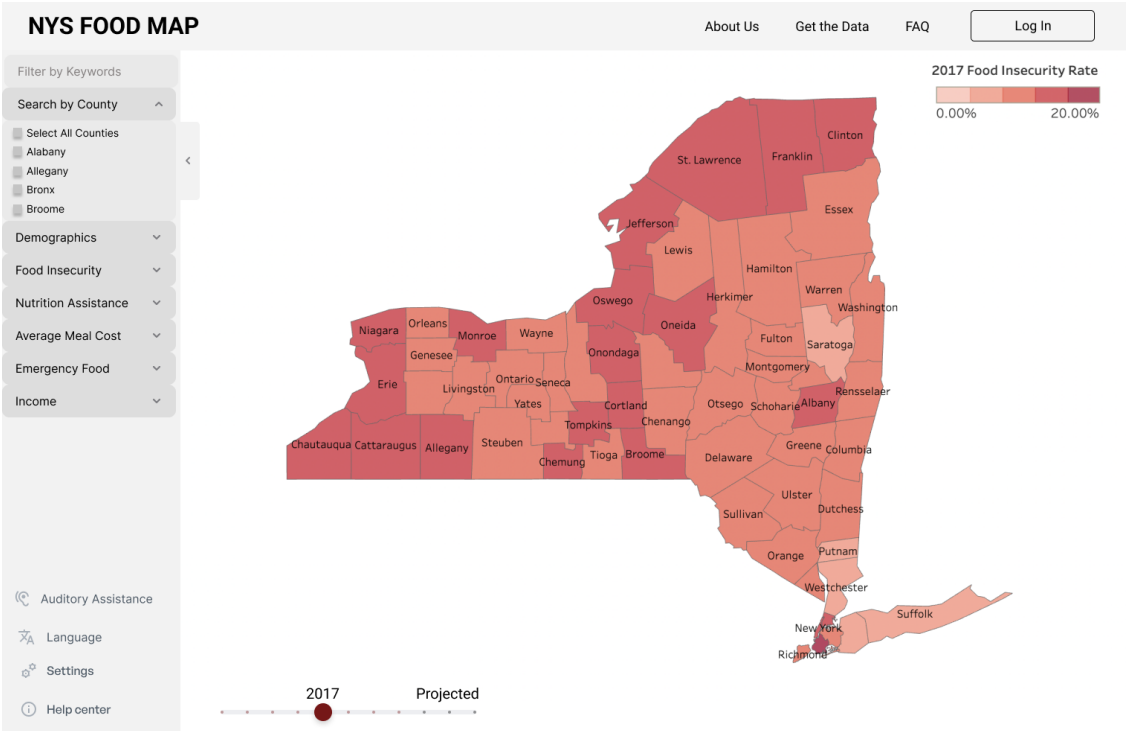


Figure 6: Food Insecurity Rate Map for 2017

The slider on the bottom left of the application allows users to select the year they want to map. Figure 6 displays FIR in 2017. If a user slides the red slider to the right, it will display the map for FIR in 2018 (Figure 7). The map is designed to be visually informative, so the contrast in color by county allows users to compare FIRs across counties visually. The inclusion of the slider allows users to seamlessly navigate FIR over time so that trends are easier to identify. The mapping interface is also designed to be fully interactive. When the user hovers over a county, a pop-up screen will appear, displaying some of the most important information about food insecurity in that county (Figure 8).

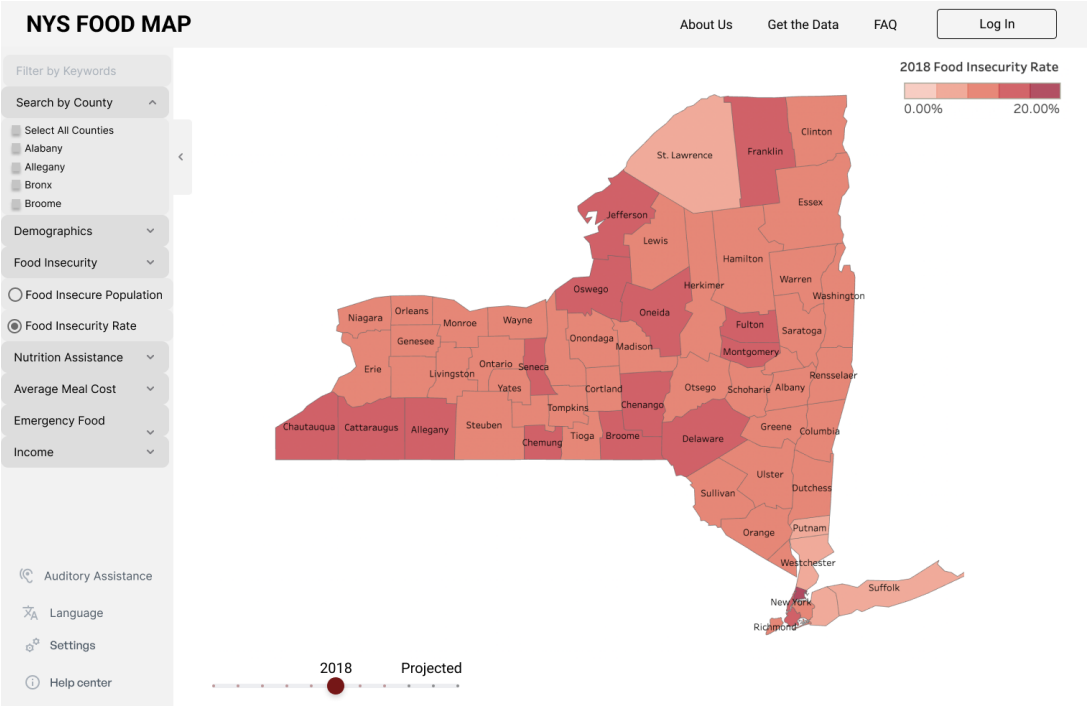


Figure 7: Food Insecurity Rate Map for 2018

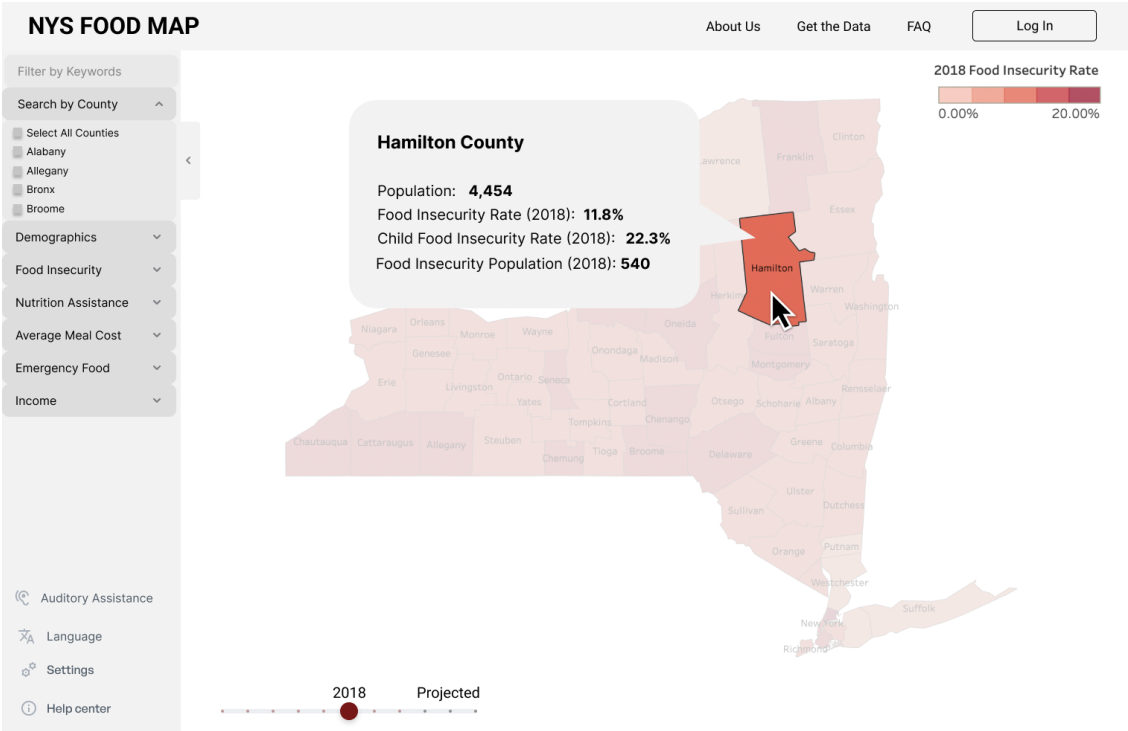


Figure 8: Example of Hovering over Hamilton County

When the user clicks on the county, the map will zoom into the selected county and display more extensive information about local food security statistics and resources (Figure 9). The design for this aspect of the site is crucial because it gives users all the information about the data and the resources available in that county. This portion of the model is designed to educate the user and direct food-insecure individuals to the appropriate resources in their area. Although not apparent through a static frame of the model, information displayed on this portal will change depending on the information, like demographics or nutrition assistance, that users chose to select on the left side of the page. All sources of information will be cited and shown on the bottom right corner of the user's screen. This creates transparency in the data used and allows users to look into the data for themselves.

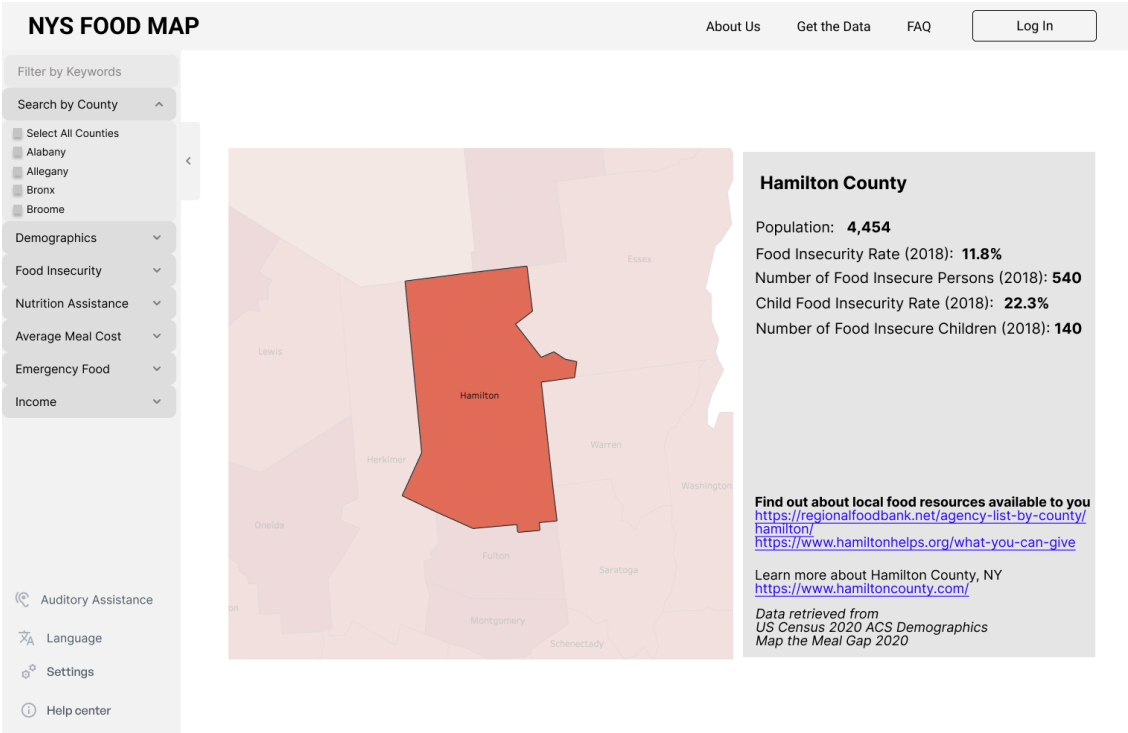


Figure 9: Example of a Selected County

One of the last functions developed for this model is the functionality to switch between a map displaying FIR and food-insecure populations (Figure 10). Users can quickly switch between the data displayed using a toggle function on the left panel. The information gives users a new perspective on food insecurity across counties. Based on the color distribution in Figure 7, users might identify Kings, Franklin, or Allegany counties as areas of concern since they have some of the highest FIRs in the state. This would mean that a higher proportion of the population in those counties experienced food insecurity. However, based on the color distribution in Figure 10, areas like Suffolk, Richmond, and Erie counties are much darker than others. This signifies that more people in those counties experience food insecurity. Depending on the user's perspective, where resources should be prioritized may differ. This tool is designed to allow users to learn more about food insecurity through a visual tool and develop an informed perspective on the issue.

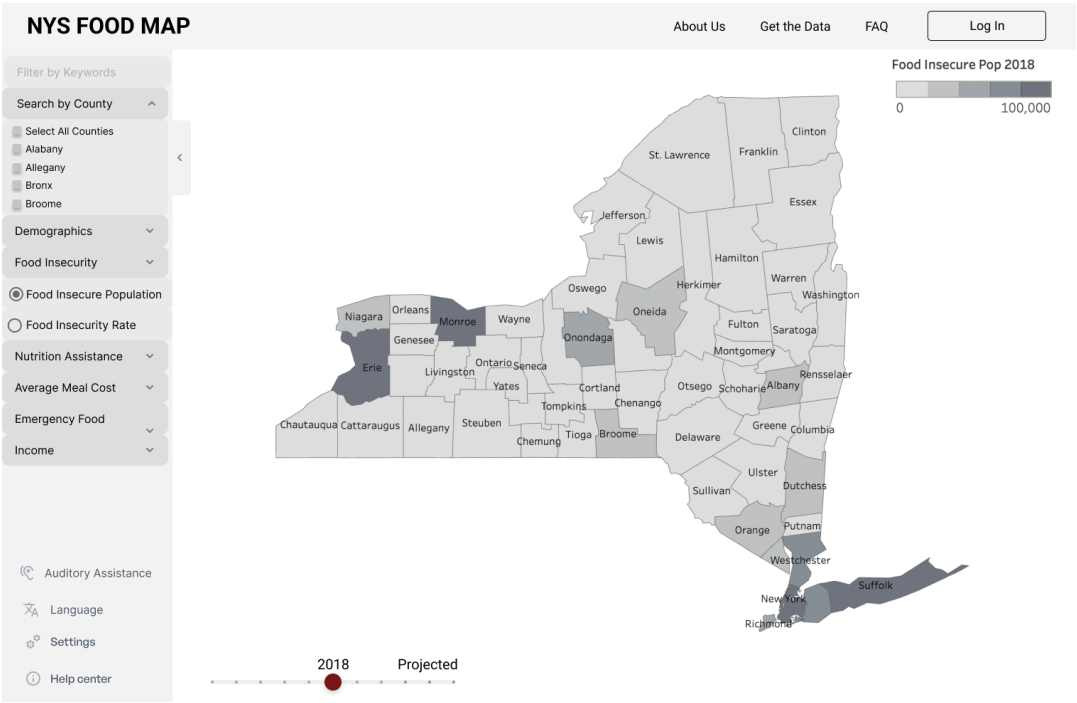


Figure 10: Mapping Food Insecure Population in 2018

**Limitations of the model**

The data used to create the NYS Food Map proposal visualization is derived from the Map the Meal Gap 2020 data. Hence, the visual information for FIR conveyed to users is similar to the existing MMG model. The limitations of the data discussed earlier still apply to this model. Most importantly, the data does not weigh the impact of FIR in minority populations other than Black and Hispanic populations. As discussed previously, to fill the knowledge gap, more work has to be conducted to obtain sufficient data to create an accurate model. If the data is incorporated into the proposed model, we will achieve a more accurate and nuanced representation of food insecurity in NYS.

**Future Improvements for the Model**

If the NYS Food Map is adopted, more work should be done to improve its functionality. An ideal function of the model would be to accurately project FIRs based on interventions implemented by the county over time. This function aims to allow organizations and government agencies to project the most effective ways to approach food security interventions to improve overall food access across the state. Policymakers can then propose informed food policies that are grounded in data. If the model proves to be accurate and helpful to the state and local stakeholders, this application can be scaled and applied to other states across the country. With this tool, stakeholders will have access to local data that helps inform appropriate interventions that best suit the community.

**Conclusion**

Ultimately, more research must be conducted for comprehensive tools for food security to be built. Many non-governmental organizations and government agencies work on projects and interventions to increase food security across the country. Witnessing many local initiatives that are passionate about serving their communities inspired me to create a single educational tool that merged data and resources. The goal of the project was to propose an accessible food security model that is easy to use and informative for all users. This project shows that food security is complicated to define and measure. Identifying specific causes of food insecurity also proved challenging, as food insecurity is a symptom of many systemic issues one may experience. Regardless of the challenges, it is crucial to simplify complicated data into digestible nuggets of information to break the barriers to accessing knowledge.

**Appendix**

Survey Questions Used by USDA to Assess Household Food Security

1. "We worried whether our food would run out before we got money to buy more." Was that often, sometimes, or never true for you in the last 12 months?

2. "The food that we bought just didn't last and we didn't have money to get more." Was that often, sometimes, or never true for you in the last 12 months?

3. "We couldn't afford to eat balanced meals." Was that often, sometimes, or never true for you in the last 12 months?

4. In the last 12 months, did you or other adults in the household ever cut the size of your meals or skip meals because there wasn't enough money for food? (Yes/No)

5. (If yes to question 4) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?

6. In the last 12 months, did you ever eat less than you felt you should because there wasn't enough money for food? (Yes/No)

7. In the last 12 months, were you ever hungry, but didn't eat, because there wasn't enough money for food? (Yes/No)

8. In the last 12 months, did you lose weight because there wasn't enough money for food? (Yes/No)

9. In the last 12 months did you or other adults in your household ever not eat for a whole day because there wasn't enough money for food? (Yes/No)

10. (If yes to question 9) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?

**(Questions 11-18 were asked only if the household included children age 0-17)**

11. "We relied on only a few kinds of low-cost food to feed our children because we were running out of money to buy food." Was that often, sometimes, or never true for you in the last 12 months?

12. "We couldn't feed our children a balanced meal, because we couldn't afford that." Was that often, sometimes, or never true for you in the last 12 months?

13. "The children were not eating enough because we just couldn't afford enough food." Was that often, sometimes, or never true for you in the last 12 months?

14. In the last 12 months, did you ever cut the size of any of the children's meals because there wasn't enough money for food? (Yes/No)

15. In the last 12 months, were the children ever hungry but you just couldn't afford more food? (Yes/No)

16. In the last 12 months, did any of the children ever skip a meal because there wasn't enough money for food? (Yes/No)

17. (If yes to question 16) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?

18. In the last 12 months did any of the children ever not eat for a whole day because there wasn't enough money for food? (Yes/No)

Source: (USDA ERS 2022)

References

Clay, Lauren A., and Stephanie Rogus. 2021. "Impact of Employment, Essential Work, and Risk Factors on Food Access during the COVID-19 Pandemic in New York State." *International Journal of Environmental Research and Public Health* 18, no. 4 (February): 12. https://doi.org/10.3390/ijerph18041451.

Coleman-Jensen, Alisha, Matthew P. Rabbitt, Christian A. Gregory, and Anita Singh. 2021. "Household Food Security in the United States in 2020." USDA ERS. Economic Research Report. https://www.ers.usda.gov/webdocs/publications/102076/err-298.pdf.

Feeding America. n.d. "Overall (all ages) Hunger & Poverty in the United States | Map the Meal Gap." Map the Meal Gap. Accessed March 7, 2023. https://map.feedingamerica.org/county/2020/overall/.

FRED. n.d. "Unemployment Rate." FRED. Accessed March 7, 2023. https://fred.stlouisfed.org/series/UNRATE.

Hake, Monica, Emily Engelhard, and Adam Dewey. 2022. "Map The Meal Gap 2022." Feeding America. https://www.feedingamerica.org/sites/default/files/2022-09/Map%20the%20Meal%20Gap%202022%20Report_0.pdf.

Hunger Solutions New York. 2023. "About Us - About Us." Hunger Solutions New York. https://hungersolutionsny.org/about-us/.

Hunger Solutions New York. n.d. "Food Insecurity and Federal Nutrition Program Participation." Hunger Solutions New York. Accessed March 7, 2023. https://hungersolutionsny.org/resources-action/hunger-in-nys/food-insecurity-and-federal-nutrition-program-participation/.

Kaufman, Phillip, Chris Dicken, and Ryan Williams. 2014. "Measuring Access to Healthful, Affordable Food in American Indian and Alaska Native Tribal Areas." USDA ERS. https://www.ers.usda.gov/webdocs/publications/43905/49690_eib131_errata.pdf.

Maillacheruvu, Sara U. 2022. "The Historical Determinants of Food Insecurity in Native Communities." Center on Budget and Policy Priorities. https://www.cbpp.org/research/food-assistance/the-historical-determinants-of-food-insecurity-in-native-communities.

Map the Meal Gap. 2022. "Map the Meal Gap 2022 Technical Brief." https://www.feedingamerica.org/sites/default/files/2022-08/Map%20the%20Meal%20Gap%202022%20Technical%20Brief.pdf.

USDA. 2022. "Measurement." USDA ERS. https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-u-s/measurement/.

USDA ERS. 2022. "The prevalence of food insecurity in 2021 is unchanged from 2020." USDA ERS. https://www.ers.usda.gov/data-products/chart-gallery/gallery/chart-detail/?chartId=58378.

USDA ERS. 2022. "What is Food Security?" USDA ERS. https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-u-s/measurement/.